

Practical 5: Potential Energy Scan of Molecules

Aim

To scan the potential energy scan of a molecule using a quantum mechanical method.

Theory:

The potential energy curve (PEC) or surface is the plot of electronic energy as a variation of nuclear geometry. The origin of PEC is due to heavy mass of nuclei, the electron move much faster than nuclei, as a result we may separate the electronic and nuclei motion – Born-Oppenheimer Approximation. This is the fundamental theory in the area of molecular dynamics. The PEC is also considered as a tool for visualization of chemical reaction. It is assumed that molecule undergoes to chemical reaction by moving over the PEC.

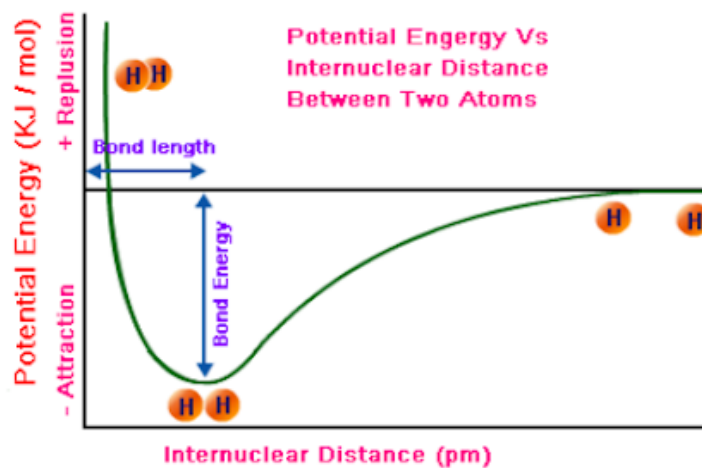


Figure: General Potential energy surface and Potential energy curve

In this session, we will generate the PEC of O₂ molecule by varying its bond distance.

Procedure:

- Build **methanol** molecule in ChemCompute.

- Set the initial **C-O bond distance** to **0.7 Å**.
- Prepare the input for single point energy calculation. Choose an appropriate method and basis set.
- Change the C-O bond distance at intervals of **0.1 Å** and repeat the calculation.
- Repeat the procedure for **15 steps** by changing the distance by **0.1 Å** in each step.
- Record the Electronic Energy (above the molecule box) and the bond distance

Results:

Draw methanol molecule and generate PEC for C-O bond.

S. No.	Bond Distance (Å)	Electronic Energy (Hartree)	Relative Energy (Hartree)
1	0.7	-303279.395	351.783
2	0.8	-303527.54	103.638
3	0.9	-303618.185	12.993
4	1.0	-303629.066	2.112
5	1.1	-303600.646	30.534
6	1.2	-303554.213	77.048
7	1.3	-303628.338	2.84
8	1.4	-303631.178	0
9	1.5	-303395.226	235.952
10	1.6	-303346.865	284.313
11	1.7	-303302.417	328.761
12	1.8	-303261.985	369.193
13	1.9	-303225.473	405.705
14	2.0	-303192.695	438.483
15	2.1	-303163.39	467.788

Figure: Relative energy vs bond distance potential energy surface

